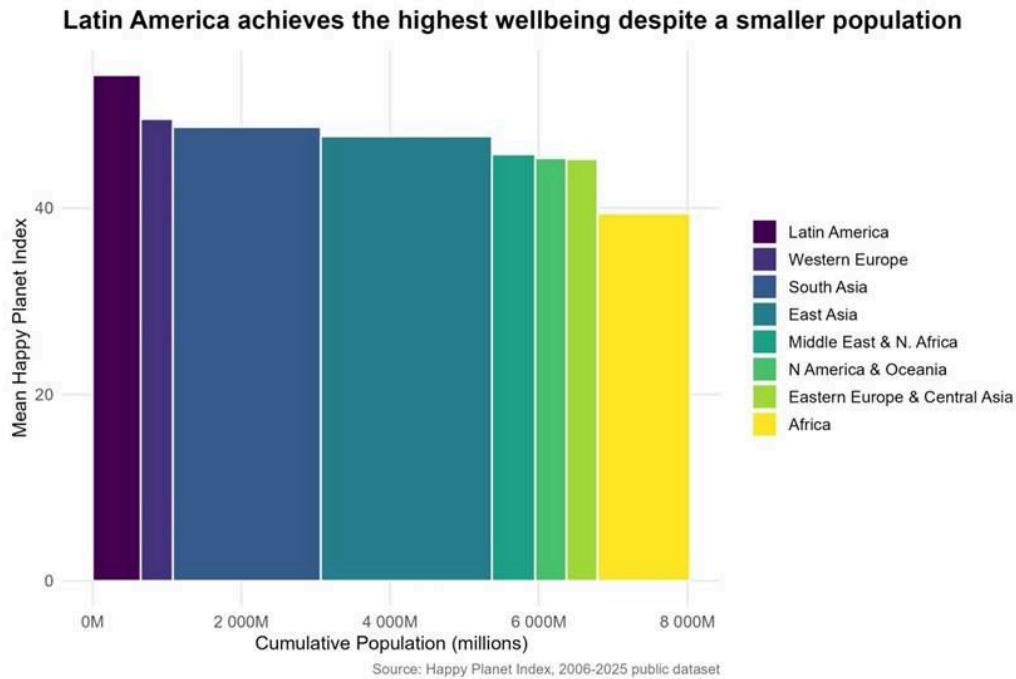


## Assignment 4: 48-Hour Chart Hackathon

Team: Md Eyasin Ul Islam Pavel, Muhammad Al Hasib, Susmita Sarkar

Chart 1: Variable-width Column Chart (HPI by Continent)



**Figure 1.** Mean Happy Planet Index by continent, with bar width proportional to population (2006-2025)

This variable-width column chart compares mean Happy Planet Index (HPI) scores across eight world regions, using data from the Happy Planet Index 2006-2025 public dataset. Bar height represents each region's average HPI score; bar width represents its population (most recent available year per country), so the chart simultaneously shows wellbeing performance and population scale. Regions are ordered left to right by descending HPI.

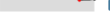
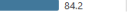
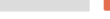
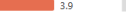
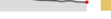
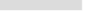
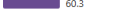
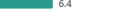
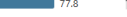
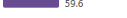
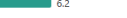
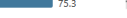
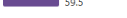
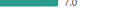
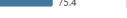
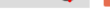
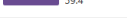
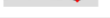
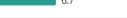
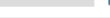
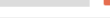
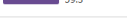
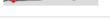
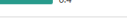
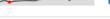
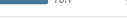
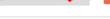
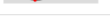
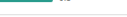
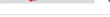
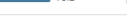
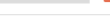
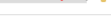
Latin America records the highest mean HPI (54.3) despite having one of the smaller populations among the eight regions, while Africa records the lowest (39.4) despite the second-largest population. This pattern suggests population size and HPI performance are not strongly related, countries achieve high sustainable wellbeing independent of how populous their region is.

Tool used: Claude (Anthropic), Sonnet 4.6

Prompts used (in order):

- "Update the codes and find any error before running in R."
- "Update the code with axis formatting, legend ordering"

Chart 2: Table with embedded charts

| Country     | HPI Rank | HPI                                                                                    |                                                                                   | Life Satisfaction (0-10)                                                              |                                                                                   | Life Expectancy (yrs)                                                                  |                                                                                    | Ecological Footprint (gha)                                                              |                                                                                     | GDP per capita (USD)                                                                       |                                                                                     |
|-------------|----------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|             |          | 2025 Performance                                                                       | Trend 2006-25                                                                     | 2025 Performance                                                                      | Trend 2006-25                                                                     | 2025 Performance                                                                       | Trend 2006-25                                                                      | 2025 Performance                                                                        | Trend 2006-25                                                                       | 2025 Performance                                                                           | Trend 2006-25                                                                       |
| Costa Rica  | 1        |  68.3 |  |  7.6 |  |  81.2 |  |  2.6 |  |  26,973 |  |
| Guatemala   | 2        |  60.7 |  |  6.7 |  |  72.9 |  |  1.8 |  |  12,641 |  |
| Tajikistan  | 3        |  60.7 |  |  6.0 |  |  72.1 |  |  1.0 |  |  4,756  |  |
| Spain       | 4        |  60.6 |  |  6.6 |  |  84.0 |  |  3.2 |  |  48,460 |  |
| Colombia    | 5        |  60.6 |  |  6.0 |  |  78.1 |  |  1.8 |  |  18,477 |  |
| Switzerland | 6        |  60.4 |  |  7.1 |  |  84.2 |  |  3.9 |  |  82,286 |  |
| Ecuador     | 7        |  60.3 |  |  6.4 |  |  77.8 |  |  2.3 |  |  13,936 |  |
| Nicaragua   | 8        |  59.6 |  |  6.2 |  |  75.3 |  |  1.8 |  |  7,662  |  |
| Mexico      | 9        |  59.5 |  |  7.0 |  |  75.4 |  |  2.6 |  |  22,040 |  |
| Brazil      | 10       |  59.4 |  |  6.7 |  |  76.2 |  |  2.4 |  |  19,652 |  |
| Philippines | 11       |  59.3 |  |  6.4 |  |  70.1 |  |  1.4 |  |  10,376 |  |
| Honduras    | 12       |  58.9 |  |  6.3 |  |  73.2 |  |  1.8 |  |  6,586  |  |
| Laos        | 13       |  58.8 |  |  5.9 |  |  69.5 |  |  0.9 |  |  8,600  |  |
| El Salvador | 14       |  58.5 |  |  6.8 |  |  72.5 |  |  2.1 |  |  11,669 |  |
| Albania     | 15       |  58.3 |  |  6.0 |  |  80.0 |  |  2.4 |  |  21,641 |  |

Source: Happy Planet Index 2006-2025 Public Dataset

Figure 1. Comparative Overview of the Top 15 Countries by HPI Rank

This Happy Planet Index (HPI) table with embedded charts shows the top 15 HPI-ranked countries and combines multiple indicators into a compact comparison. I have selected all the indicators for the chart.

The table is ordered by country based on its 2025 HPI rank. The table shows the country-wise performance in HPI, life satisfaction, life expectancy, ecological footprint, and GDP per capita. The latest value for each indicator is shown with inline bars for comparison, and the sparklines show the historical trend over time, thus showing how each indicator has changed across the HPI years. Instead of separate charts for each indicator, the inline bars and sparklines provide country-wise comparisons and temporal patterns in a single chart.

**AI Tool for chart preparation:** claude.ai

**Model:** Sonnet 5.5

**AI Prompts used:**

Prompt 1: I need to prepare a Table with embedded charts I want to prepare it for many countries and 3 categories. Suggested HPI design - Countries × indicators with inline bars or sparkline. Suggested R Approach: gt + gtExtras, or facet\_wrap()

Prompt 2: Using the table I already built for all countries (gt with inline bars and sparklines for the HPI categories), modify the code to display only the 15 countries with the best HPI rank in 2025 (rank 1 to 15). Keep the same columns, bars, sparklines and styling, sort by HPI rank, and update the title "Top 15 countries by HPI rank".

Prompt 3: Modify the existing HPI table to improve its labeling and presentation. Remove the “cat1,” “cat2,” etc. prefixes from the indicator headings and use the actual indicator names: HPI,

Life Satisfaction (0–10), Life Expectancy (yrs), Ecological Footprint (gha), and GDP per capita (USD). Rename the “latest” columns to “2025 Performance” and the “trend” columns to “Trend 2006–25” while keeping the existing inline bars and sparklines. Add a small, right-aligned source note below the table: “Source: Happy Planet Index 2006–2025 Public Dataset.”

**Chart 3: Costa Rica leads the top 20 countries, while Mongolia ranks lowest among the bottom 20**

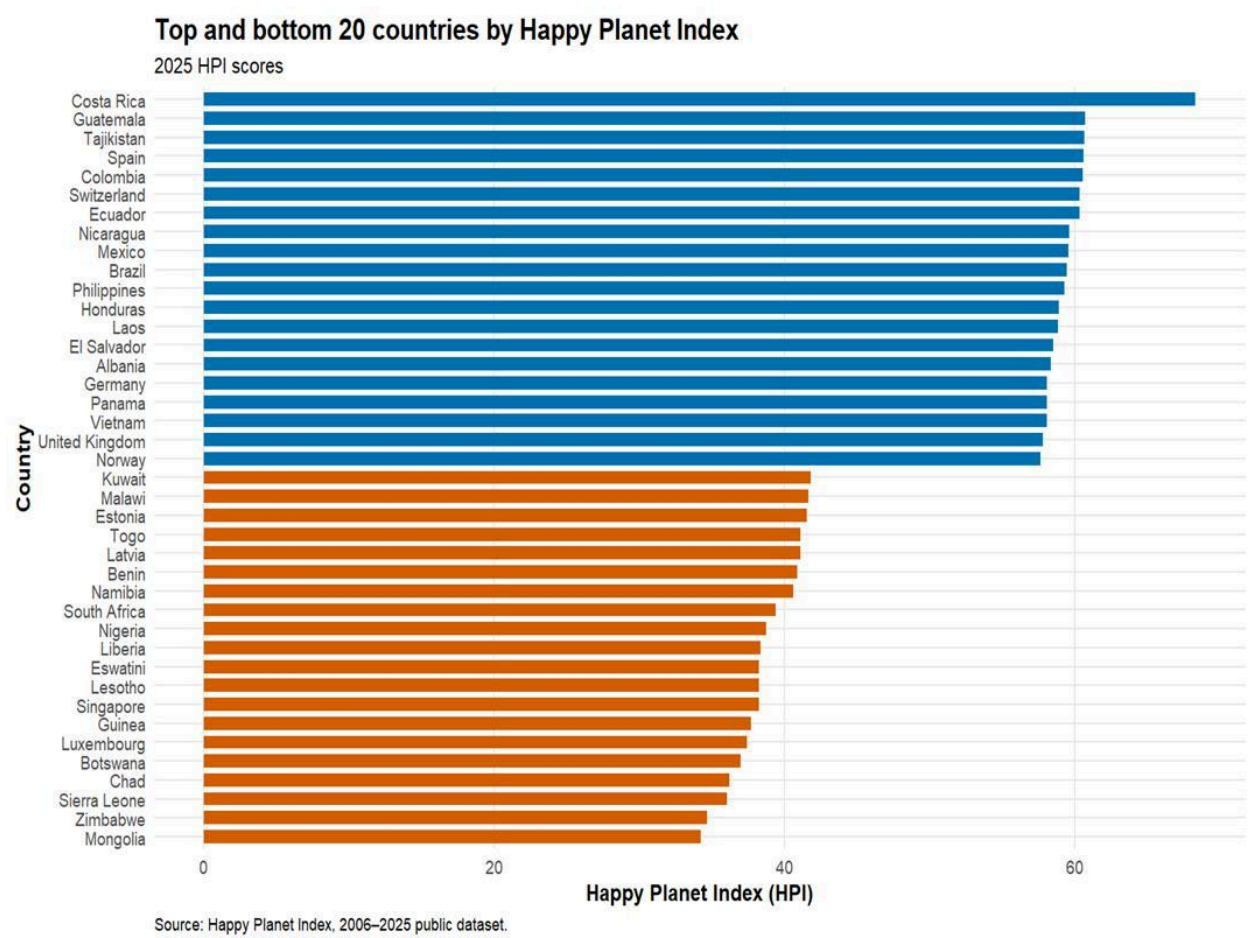


Chart 3 compares the 20 countries with the highest and lowest Happy Planet Index (HPI) scores in 2025. Costa Rica has the highest HPI score among the countries shown, while Mongolia has the lowest, highlighting substantial differences in HPI across countries.

**AI Tool:** ChatGPT  
**Model:** GPT-5.6 Luna  
**Date:** October 2, 2026

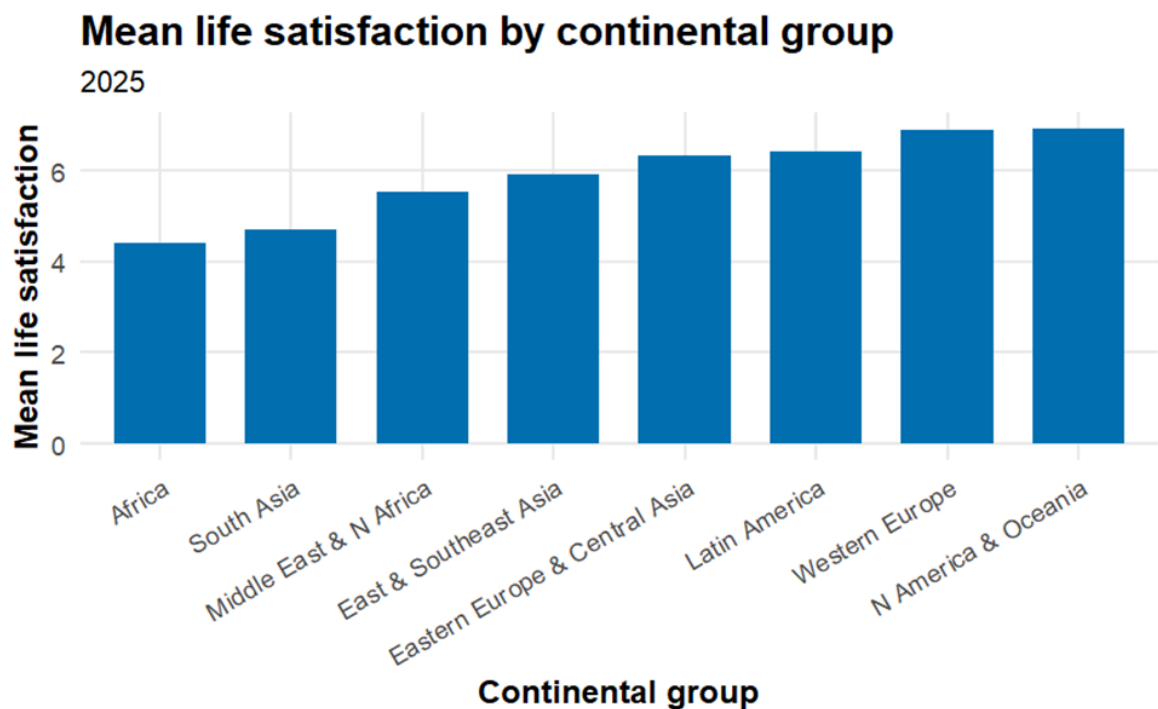
**Prompt:**

I am working on EPPS6356 Data Visualization Assignment 4 using the Happy Planet Index dataset. Help me create Chart 3 in R using ggplot2. Chart 3 should be a bar chart showing the top 20 and bottom 20 countries by Happy Planet Index (HPI) score for 2025, sorted by HPI. Use a color-blind-safe color palette, clearly label the country and HPI axes, include a takeaway-oriented title, and include a source caption. Please provide reproducible R code that I can run from the HPI Excel dataset.

**Discussion/Fixes:**

ChatGPT provided R code using dplyr, ggplot2, and forcats to identify and order the top and bottom 20 countries by HPI. I adjusted the file path to match the location of the dataset on my computer and verified that the chart rendered correctly. I also used separate colors for the top and bottom groups and adjusted the theme and labels for readability.

**Chart 4: Mean life satisfaction is highest in N America & Oceania and lowest in Africa**



Source: Happy Planet Index, 2006–2025 public dataset.

Chart 4 shows the average life satisfaction score for each continental group in 2025. N America & Oceania has the highest mean life satisfaction, while Africa has the lowest among the continental groups shown.

**AI Tool:** ChatGPT  
**Model:** GPT-5.6 Luna  
**Date:** October 2, 2026

**Prompt:**

I am working on EPPS6356 Data Visualization Assignment 4 using the Happy Planet Index dataset. Help me create Chart 4 in R using ggplot2. Chart 4 should show the mean Life Satisfaction score by continental group for 2025. Calculate the mean Life Satisfaction for each continental group, create a column chart, sort the groups by their mean value, use a color-blind-safe color, clearly label both axes, include a takeaway-oriented title, and include a source caption. Please provide reproducible R code using the HPI Excel dataset.

**Discussion/Fixes:**

ChatGPT provided R code to calculate mean Life Satisfaction by continental group and create the column chart using ggplot2. I adjusted the dataset file path and mapped the dataset's continent codes to their corresponding continental group names. I also adjusted the axis labels, title, rotation of the x-axis labels, and overall theme to improve readability.

## **Team Synergy Report**

Our team has three members: Md Eyasin Ul Islam Pavel, Muhammad Al Hasib, and Susmita Sarkar. We divided the visualization work among team members, so each person developed specific charts, and Susmita Sarkar compiled the final assignment.

Md Eyasin Ul Islam Pavel was responsible for the development of Chart 3 and Chart 4, . Muhammad Al Hasib was responsible for Chart 1, Susmita Sarkar was responsible for Chart 2 as well as compilation, and presentation of all four charts.